

ITA 1412 - Ethical Hacking R. Pravalika
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① Nmap Scanning without Permission Ethical and legal Analysis.

Nmap is a network scanning tool used to identify open ports, services, and security weaknesses. It is commonly used by security professionals for authorized security testing.

However, using Nmap without permission is considered unethical and may be illegal.

1. Ethical Analysis

- > The student performed the scan without the company's permission.
- > Even though no data was stolen, systems were damaged.

2. legal Analysis under the IT Act, 2000

Relevant provisions include.

> Section 43:- Un authorized access or interference with computer systems may result in compensation.

> Section 66:- If unauthorized access is done dishonestly or fraudulently, it becomes a criminal offense.

② SQL injection vulnerability - Ethical and legal Analysis.

SQL injection is a cyber attack in which an attacker inserts malicious SQL commands into a website's input fields. In this case, the researcher has publicly disclosed.

Ethical Analysis

- The researchers acted unethically by publicizing the vulnerability immediately
- Ethical cybersecurity practice requires responsible disclosure
- Public disclosure before fixing the issue exposed customers to serious risk
- legal analysis under the IT Act, 2008

Section 43: Unauthorized access or damage to computer systems.

Section 66 - Unauthorized access performed dishonestly or fraudulently

Section 72 - Unauthorized disclosure of confidential

cybercriminal who exploited the vulnerability.

③ Phishing Attack - Ethical and legal Analysis.

The attack described is a phishing Attack. Phishing is a cyber attack where attackers send fake emails, messages or websites to trick users into revealing sensitive information.

1. The attack is phishing
2. The attacker impersonated the company's IT support team
3. The employee entered login
4. The attacker then used these credentials to gain unauthorized access to confidential company data.

Section 43 - unauthorized access or damage to computer systems can result in compensation.

- Provide regular cybersecurity awareness training.
- Implement Multi-factor Authentication
- Install email filtering and anti-phishing solutions.
- Monitor network activity and respond quickly to incidents.

④

Ethical Hacking vs Unauthorized Hacking
Ethical and Legal Analysis.

A cybersecurity student discovered a security vulnerability in an online banking portal. To prove the seriousness of the issue.

The student accessed a small amount of customer information without information.

- The student had good intentions but acted without permission.
- Accessing customer information without violates
- Ethical hackers must always obtain prior written permission before testing any system

- The student's actions.

Ethical
Hacking

Unauthorized

- Done with
Permission

- Done without permission

- Legal and
Ethical

- illegal and unethical

- Improves
System
Security

- May cause harm
or data theft